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Male dog urination posture preference: A survey of owner observations



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ABSTRACT

Some mature male dogs often or almost always raise a hind limb to urinate, whereas others do so rarely or never. Previously, the largest studies exploring this behavior have focused on observations made in animal shelters. This study sought to use a citizen science approach to enable owners to report their observations of their male dogs in the home environment. A survey was devised using Google Forms and distributed via social media (Facebook, Reddit) asking owners to report the castration status, age, urination posture preference, and geographical region of their male dog(s). The survey was open from 31 March 2020 until 7 April 2020 and received 120 responses. Results were tabulated and analyzed in Google Sheets and using online statistical calculators. Approximately 76% of dogs were castrated, with contributors located in Australia/New Zealand (24%), Europe/the United Kingdom (22%), Asia (22%), and North America (32%). Many owners reported that their dog never raised a leg to urinate ($n = 36$, 30%) or had no clear preference (ambilaterality) for right or left hind limb during urination ($n = 20$, 17%), whereas other owners reported a moderate-to-strong preference for raising the left hind limb ($n = 21$, 18%) or the right hind limb ($n = 43$, 36%) during urination. Of the 36 dogs that never raised a hind limb to urinate, 33 were castrated (92%). The survey results indicated that castration was negatively associated with raised-leg urination (RLU) and that in dogs with an appreciable right or left preference for RLU, the right hind limb was raised by more male dogs than the left hind limb; however, most dogs were at least occasionally ambilateral with respect to RLU behavior. There was no statistically significant difference between age categories with respect to urination posture.

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Introduction

Laterality and ambilaterality or “pawedness” of dogs has been reported in several studies, but most studies focused on forelimb laterality and ambilaterality (equivalent to human “handedness”), with assessments based on tasks using forepaws only (Ocklenburg et al., 2019 and references therein). Typical tests of canine pawedness included the Kong™ test, tape removal test, and first-stepping test (Tomkins et al., 2010), but low numbers of study participants frequently led to conflicting or inconsistent results. A recent meta-analysis confirmed that laterality (pawedness) and ambilaterality do indeed exist in dogs with respect to the use of dogs’ forepaws (Ocklenburg et al., 2019).

Studies on the laterality and ambilaterality of canine raised-leg urination (RLU) have been previously reported in the scientific literature (Berg, 1944; Branson, 2006; Gough & McGuire, 2015; McGuire & Gough, 2017). The earlier two studies reported on low numbers of individual dogs (5 and 16 dogs, respectively). The largest and most recent studies by Gough & McGuire (2015) and McGuire & Gough (2017) focused entirely on shelter dogs and found that males were significantly more likely than females to raise a hind limb when urinating. McGuire & Gough (2017) reported that urination posture with a raised hind limb was less common in castrated males than intact males. Age influenced RLU, with propensity to show RLU increasing with age, and that most dogs were ambilateral with respect to RLU behavior. Adult and senior male dogs showed a spectrum of propensities to display RLU behavior: rarely or never (15%), occasionally to usually (22%), or almost always to always (63%) (values estimated from Fig. 2A in Gough & McGuire, 2015). In a subset of 20 male dogs with at least 20 urinations observed, 2 of 20 (10%) preferred to raise the left hind limb; 1 of 20 (5%) preferred to raise the right hind limb; and 85% were ambilateral

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(Gough & McGuire, 2015). This classification was based on stringent inclusion criteria for right or left hind limb–raising preference of an absolute Z-score >1.96 . In a subsequent study, the authors reported ambilaterality in 50% of dogs, right-preferent behavior in 19.6%, and left-preferent behavior in 30.4% of dogs with respect to RLU (McGuire & Gough, 2017).

The present study sought to examine whether dogs showing RLU were indeed predominantly ambilateral with respect to this task, based on owner observations in a nonshelter setting. This was achieved using a short “citizen science” survey distributed through social media. Null hypotheses included that there are no significant differences in mature male dog urination posture preferences based on castration status, age, or geographic location.

Methods

A survey was constructed using Google Forms, with the following questions:

1. My dog is: (choose one, required answer)
 - (a) Male (entire, not castrated).
 - (b) Male (neutered, castrated).
2. He is now approximately ___ years of age: (choose one, required answer), with options 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12 and 13+.
3. When he pees he: (choose one, required answer)
 - (a) Never raises a leg to pee – he always squats or stands on all four limbs.
 - (b) Always or almost always ($>75\%$ of occasions) raises his LEFT hind limb to pee.
 - (c) Always or almost always ($>75\%$ of occasions) raises his RIGHT hind limb to pee.
 - (d) Usually ($>50\%$ of occasions) raises his LEFT hind limb to pee.
 - (e) Usually ($>50\%$ of occasions) raises his RIGHT hind limb to pee.
 - (f) Has no clear preference regarding his posture when he pees.
4. Where in the world do you live? (choose one, not a required answer)
 - (a) Africa
 - (b) Asia
 - (c) Australia/New Zealand
 - (d) Europe/UK
 - (e) North America
 - (f) South/Central America
 - (g) Other

The survey was published initially on Facebook on 31 March 2020 and shared by 11 friends and colleagues, including 1 Thai veterinary pathologist who shared a version translated into Thai. The survey was also subsequently posted on Reddit, to the following subreddits: r/AskVet, r/CitizenScience, r/DOG, r/dogs, r/NoStupidQuestions, r/Pets, and r/SampleSize.

Responses were monitored daily, and when a total of 120 responses were received on 7 April 2020, the survey was closed. Results were exported to Google Sheets and manually converted to numerical data for statistical analysis. Initial calculations were performed in Google Sheets, and statistical tests were performed using freely available online calculators (medcalc.org; vassarstats.net for Fisher's exact test 4×2 contingency table to evaluate age category statistical differences). Statistical significance was set at $P < 0.05$.

Results

Castration status

Twenty-nine of 120 dogs (24%) were entire, and 91 dogs (76%) were castrated. Dogs from Australia/New Zealand, Europe/the

United Kingdom, and North America were very likely to be castrated (81–85%), whereas dogs from Asia were equally likely to be castrated or entire (50%). Of the 36 dogs that never raised a leg to urinate, 3 (8%) were entire and 33 (92%) were castrated. This indicates that 3 of 29 (10%) of the entire male dogs in the survey and 33 of 91 (36%) of the castrated male dogs in the survey never raise a leg to urinate, which is a statistically significant difference as per the Fisher's exact test (2×2 contingency table): exact probability = 0.0096.

Age distribution

The following age categories were assigned: <1 –2 years, >2 –5 years, >5 –9 years, and >9 years, with 29, 28, 27, and 36 individuals within each age category, respectively. In each age category, approximately 75% (range 75%–78%) of dogs were castrated.

Figure 1 shows the urination posture preferences of the dogs in this study, stratified with respect to age category. There were no statistically significant differences detected between age categories using the Freeman-Halton extension to Fisher's exact test (4×2 contingency table). Approximately half of dogs in each age category had a preference for left or right RLU, and approximately half never raised a leg or were ambilateral with respect to RLU. If a preference was reported, then approximately twice the number of dogs preferred to raise the right leg compared with those that preferred to raise the left leg, and this was consistent across all age categories.

Urination postures

There were 20 dogs that showed no clear preference (ambilateral) for which leg they raised during urination, and of those 20, 6 (30%) were entire and 14 (70%) were castrated. This indicates that 6 of 29 (21%) entire male dogs and 14 of 91 (15%) castrated male dogs had no clear preference for which leg they raised but did display leg-raising behavior during urination (ambilaterality). Castration status did not have a statistically significant influence on this category of RLU behavior (Chi-squared test for comparison of two proportions: Chi-squared = 0.443, DF = 1, $P = 0.51$).

The left leg was raised usually ($n = 5$) or almost always ($n = 16$) by 21 dogs. Of these, 6 of 29 (21%) were entire, and 15 of 91 (16%) were castrated. There was no statistically significant difference between these RLU categories based on castration status (Chi-squared test for comparison of two proportions: Chi-squared = 0.268, DF = 1, $P = 0.60$).

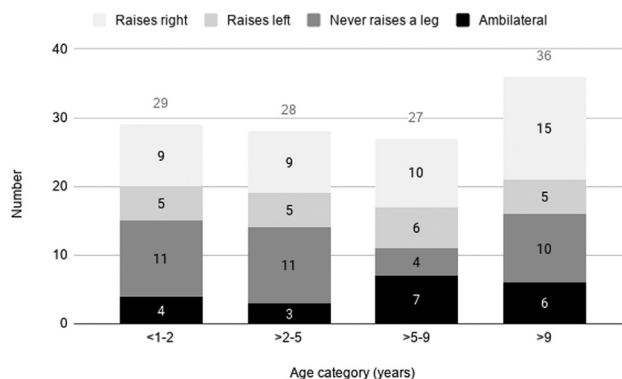


Figure 1. Male dog urination posture preferences stratified with respect to age category. Within each age category, the number of dogs that preferred to raise the right hind limb, the left hind limb, never raised a hindlimb, or were ambilateral with respect to RLU are presented. No statistically significant differences are identified using the Freeman-Halton extension to Fisher's exact test. RLU, raised-leg urination.

The right leg was raised usually ($n = 17$) or almost always ($n = 26$) by 43 dogs. Of these, 14 of 29 (48%) were entire, and 29 of 91 (32%) were castrated. There was no statistically significant difference between these RLU categories based on castration status (Chi-squared test for comparison of two proportions: Chi-squared = 2.554, $DF = 1$, $P = 0.11$).

For the 64 dogs in which owners identified a preference for raising the left or right leg during urination, the right leg ($n = 43$, 67%) was twice as likely to be raised compared with the left leg ($n = 21$, 33%), and this is a statistically significant finding (Chi-squared test for comparison of two proportions: Chi-squared = 14.117, $DF = 1$, $P = 0.0002$). Castration status did not appear to influence the degree (usually versus almost always) of the leg-raising behavior.

Geographical location

No respondents reported Africa or South/Central America as their geographic location. One contributor nominated “Other” as their geographical location, and this result was excluded from the following analysis. Sixteen of 29 dogs (55%) from Australia/New Zealand (southern hemisphere) had a preference for raising a right or left leg to urinate, and of these 16, 6 (38%) preferred to raise the right leg and 10 (62%) preferred to raise the left leg. Combining the results from the northern hemisphere respondents (Europe/the United Kingdom, North America, and Asia) yielded 90 dogs in total, of which 48 (53%) had a preference for raising a right leg or left leg to urinate. Thirty-seven of these 48 dogs (77%) preferred to raise their right leg and 11 (23%) preferred to raise their left leg. For each northern hemisphere location, raised right legs outnumbered raised left legs (by a factor of 1.4 to 14 times). Comparing southern hemisphere with northern hemisphere dogs revealed a statistically significant difference in dogs preferentially raising the left hind limb in the southern hemisphere and a right hind limb in the northern hemisphere (Fisher’s exact test (2×2 contingency table): Exact probability = 0.0057).

Discussion

This study reports on a brief survey of owner observations of their male dogs’ urination behavior, which aimed to determine whether there was any discernible owner-observed preference by dogs regarding posture during urination and to explore some factors that may influence preferred urination posture. The respondents were approached via social media, and those who elected to participate were distributed widely around the world. As such, this analysis forms a small pilot study but is not definitive. Nevertheless, some interesting and statistically significant results were obtained.

The age of the dog did not appear to appreciably influence urination posture preferences in the present study, and this is in contrast to the findings of Gough & McGuire (2015). However Gough & McGuire (2015) included dogs as young as 4 months in their study cohort, so discrepancies in the age distributions of the dogs between their study and the present study may explain this difference.

Castration was statistically significantly and negatively associated with RLU, as the vast majority of dogs that never raise a leg to urinate were castrated. In a study of the effect of testosterone on the development of urinary behavior in beagles, Ranson & Beach (1985) tentatively concluded that testosterone acting before birth and during the juvenile period prepares critical central nervous system mechanisms to enable expression of adult male urinary behavior at the appropriate stage of maturation. However, in the present study, there were 3 dogs that never raised a leg to urinate despite being entire males, so androgenic influences do not apparently guarantee

RLU behavior. Indeed, some authors suggest certain stressful stimuli may induce male dogs to temporarily revert to non-RLU postures (Berg, 1944; Martins & Valle, 1948; McGuire & Gough, 2017) and that observing the return of RLU behavior could be used to monitor adaptation to shelter environments, for example.

Any follow-up studies might investigate age at castration as a possible factor, but this was not carried out in the present study. Gough & McGuire (2015) and McGuire & Gough (2017) note that the shelter environment may influence RLU behaviors in dogs, and some dogs included in their studies had recently undergone castration, which could also conceivably impact on the observed RLU behavior in the shelter environment. It is possible that pain associated with degenerative joint disease in older dogs may negatively affect the likelihood of them expressing RLU behavior. Future studies could investigate this point.

With respect to geographical location, the only statistically significant finding in the present study was that dogs in the southern hemisphere (Australia/New Zealand) were more likely to prefer raising their left hind limb than their right hind limb compared with dogs in the northern hemisphere, in which this preference was reversed. The low numbers in these groupings demand that these results be interpreted cautiously. Left-preferent RLU behavior was identified by McGuire and Gough (2017) in their study of shelter dogs in North America (NY, USA). Perhaps, factors such as number of dogs in the household or being walked simultaneously or on which side of their body the human handler prefers to walk the dog may introduce biases in the observed RLU in some male dogs. Indeed, in a very recent study McGuire et al. (2020) identified the sex of the handler influenced mature male dog urination posture. Future studies seeking to expand on the current results, which could include a greater number of questions to identify as many factors influencing mature male RLU as possible, could be devised.

The present study suggests that with respect to male canine RLU, ambilaterality is present in 25–50% of dogs, with preferred left hind limb raising seen in 19–25% of dogs and preferred right hind limb raising seen in 20–51% of dogs, depending on the inclusion criteria in each category. The ranges result from whether mature dogs that show right or left hind limb RLU on 50–75% of occasions should be considered ambilateral or right or left preferent. The study by McGuire and Gough (2017) indicated that 23 of 41 (56%) male dogs in two North American (NY, USA) shelters were ambilateral with respect to RLU, whereas 11 of 41 (27%) preferred to raise the left hind limb compared with 7 of 41 (17%) that preferred to raise the right hind limb.

If the results of the study by McGuire and Gough (2017) are combined with the observations in the present study, for those mature male dogs that display RLU, 43–65 of 125 (34–52%) dogs are considered ambilateral, 33–50 of 125 (26–40%) dogs are right preferent, and 27–32 of 125 (22–26%) dogs are considered left preferent with respect to RLU. This is broadly in keeping with the results of a recent meta-analysis on forelimb pawedness in dogs, which found that forelimb laterality and ambilaterality exist in dogs, with approximately 37% considered ambilateral, 32% considered right preferent, and 31% considered left preferent (Ocklenburg et al., 2019).

Conclusions

This pilot study indicates that castration is negatively associated with RLU in dogs. Age (beyond puppyhood, >1 year) does not appear to be a significant factor in expression of RLU behavior. Ambilaterality and preferential raising of the left or right hind limb during urination all appear to exist in dogs, with ambilaterality most common, but where a preference exists, the preference for raising the right hind limb exceeds the preference for raising the left hind limb. Larger observational studies of pet dogs in their usual home environment are needed to investigate these findings further

and to clarify the potential influences of age at castration and other potentially confounding factors.

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Conflict of interest

The author declares no conflicts of interest.

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